

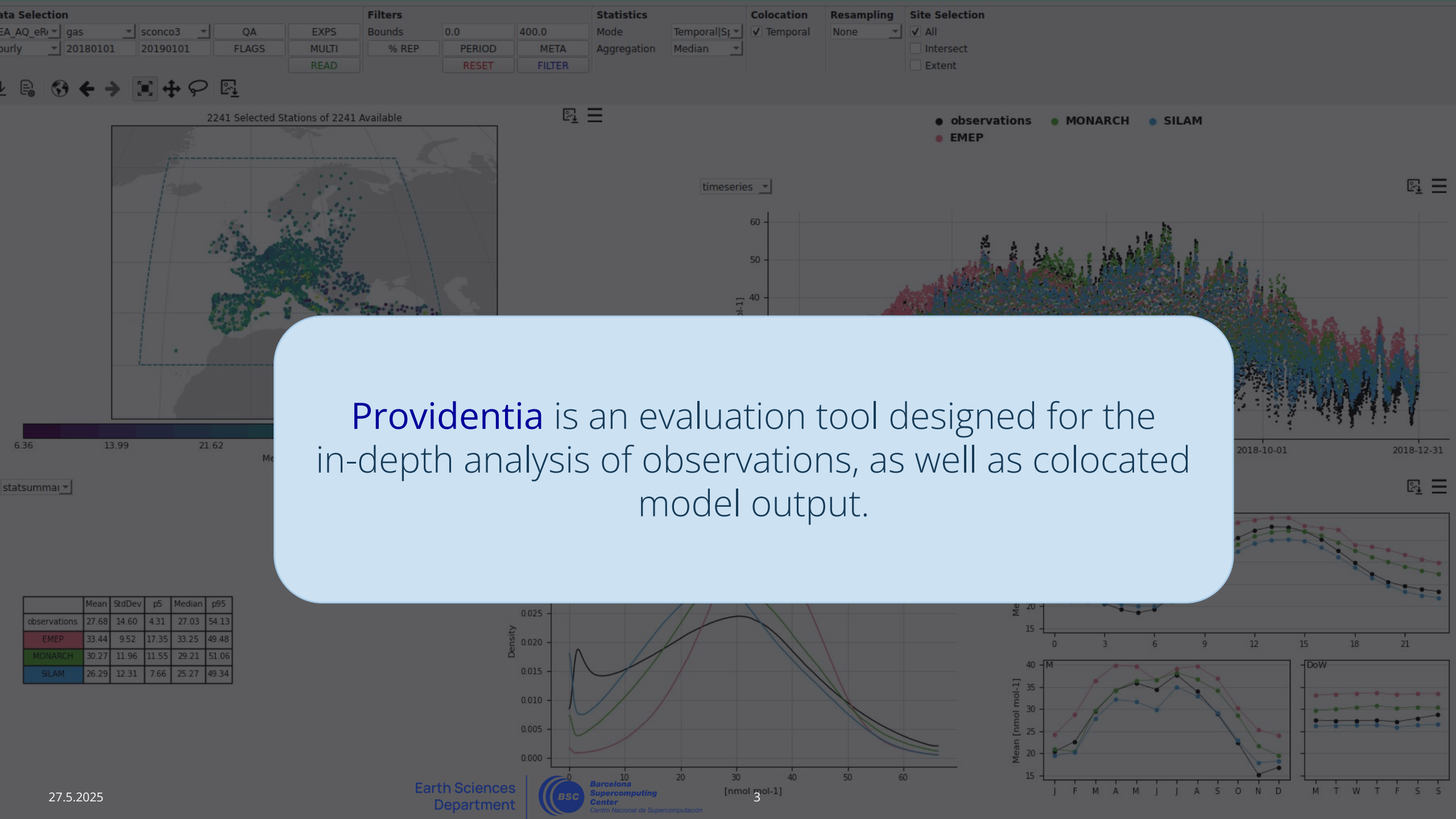
Providentia

An Evaluation and Visualisation Tool for ACTRIS Data

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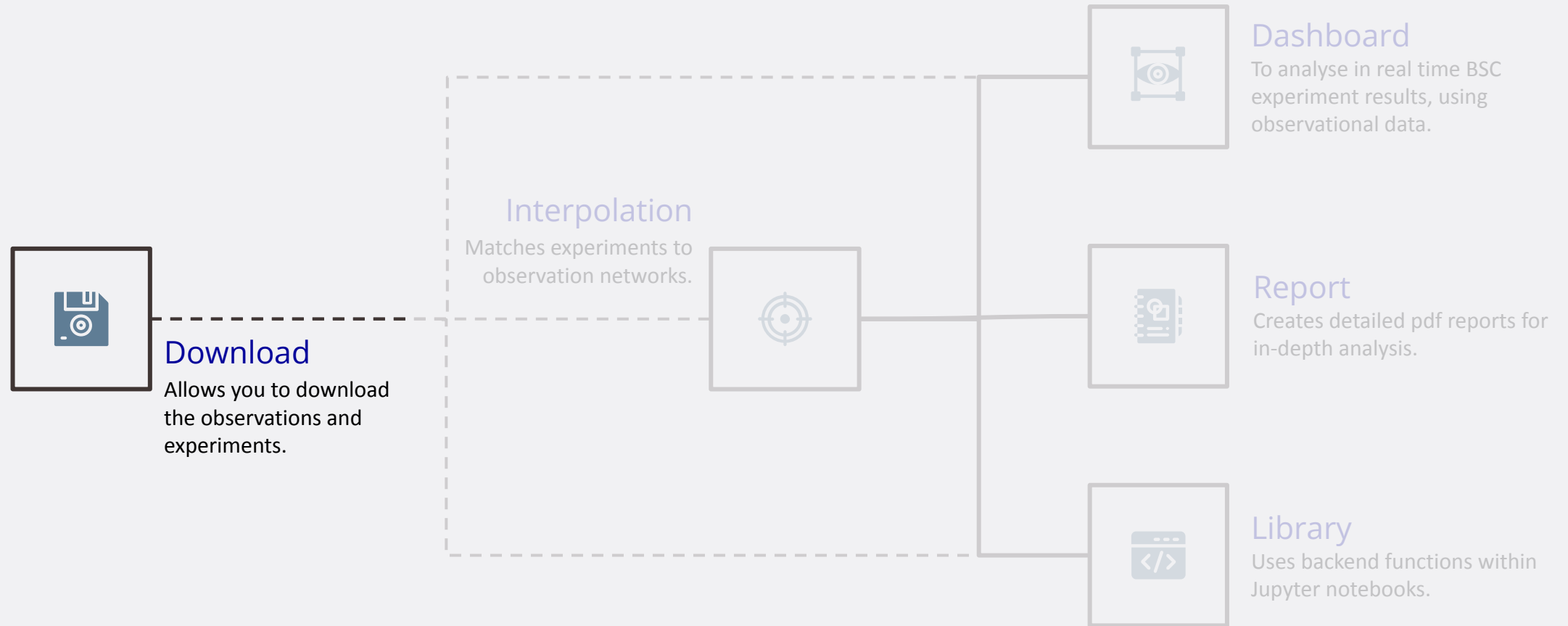
Introduction



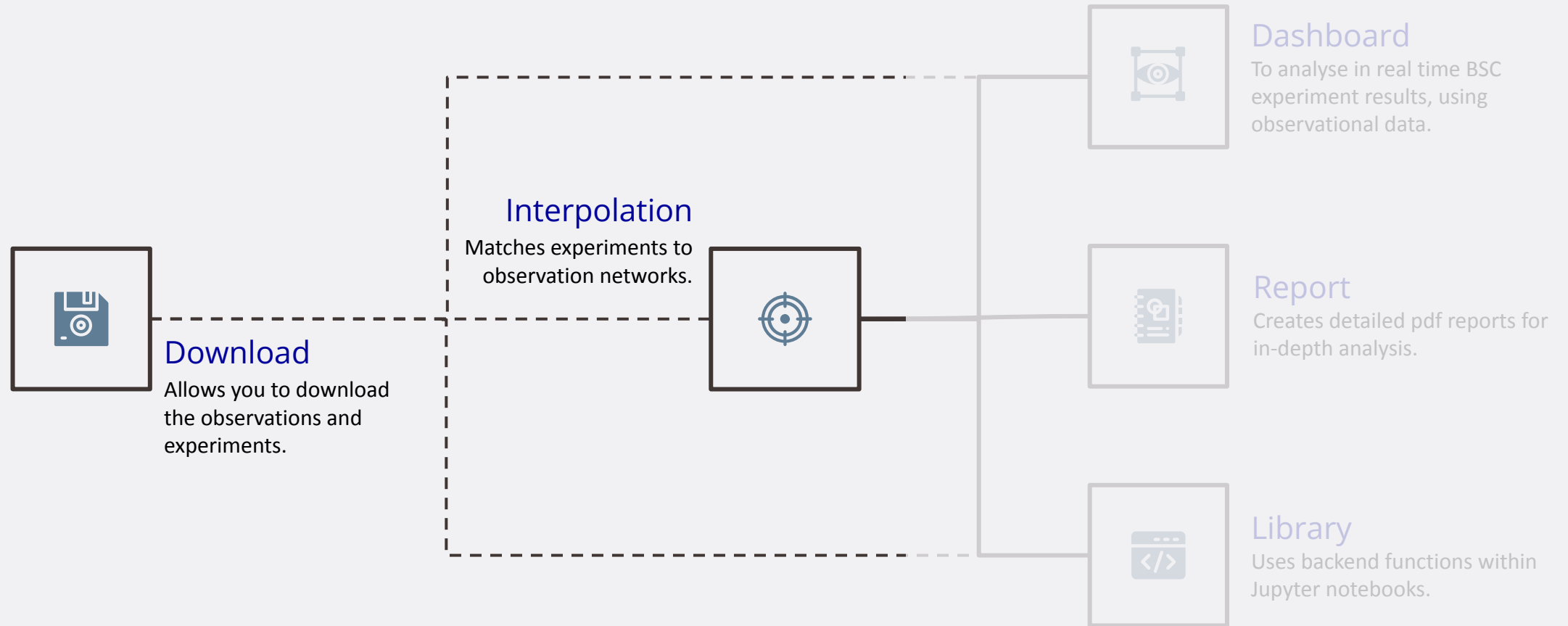
Providentia is an evaluation tool designed for the in-depth analysis of observations, as well as colocated model output.

Key Functionalities

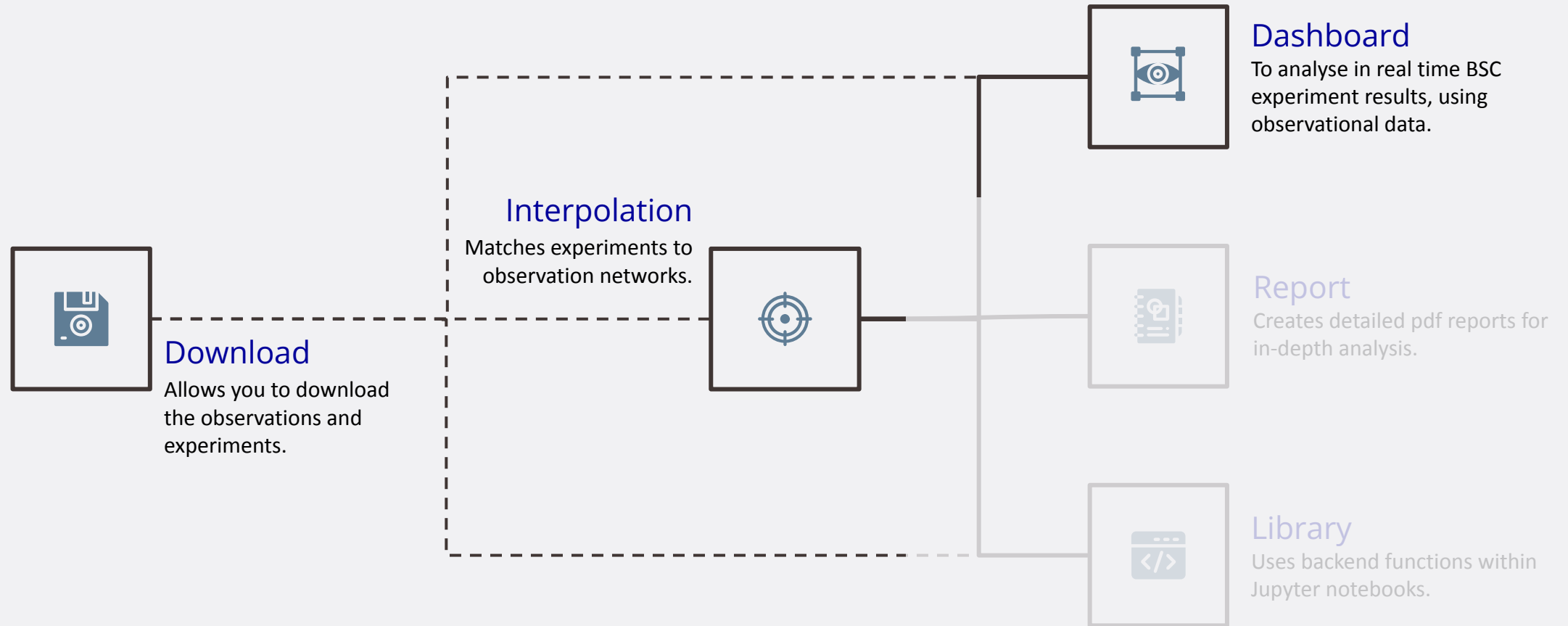
Structure



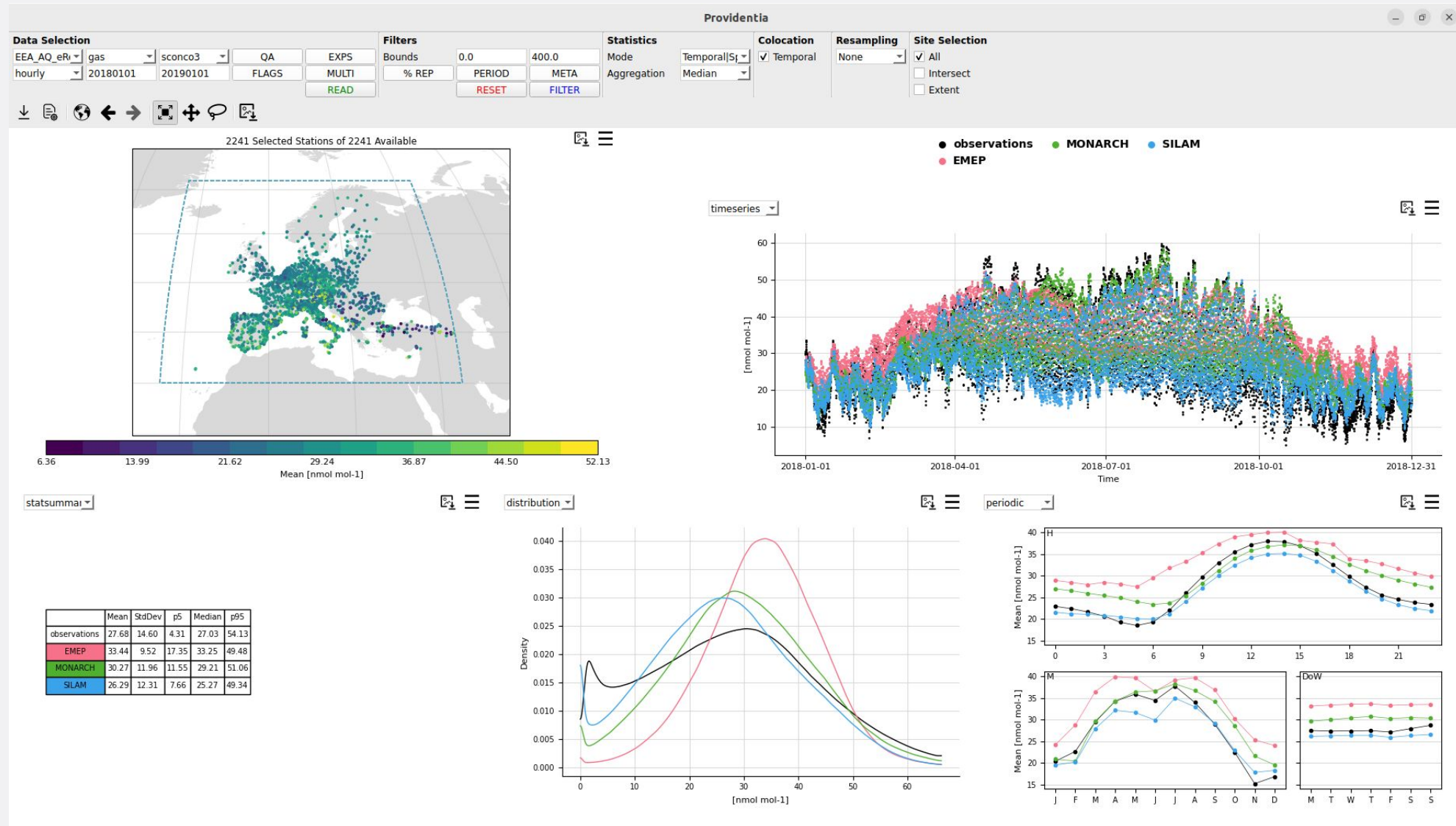
Structure



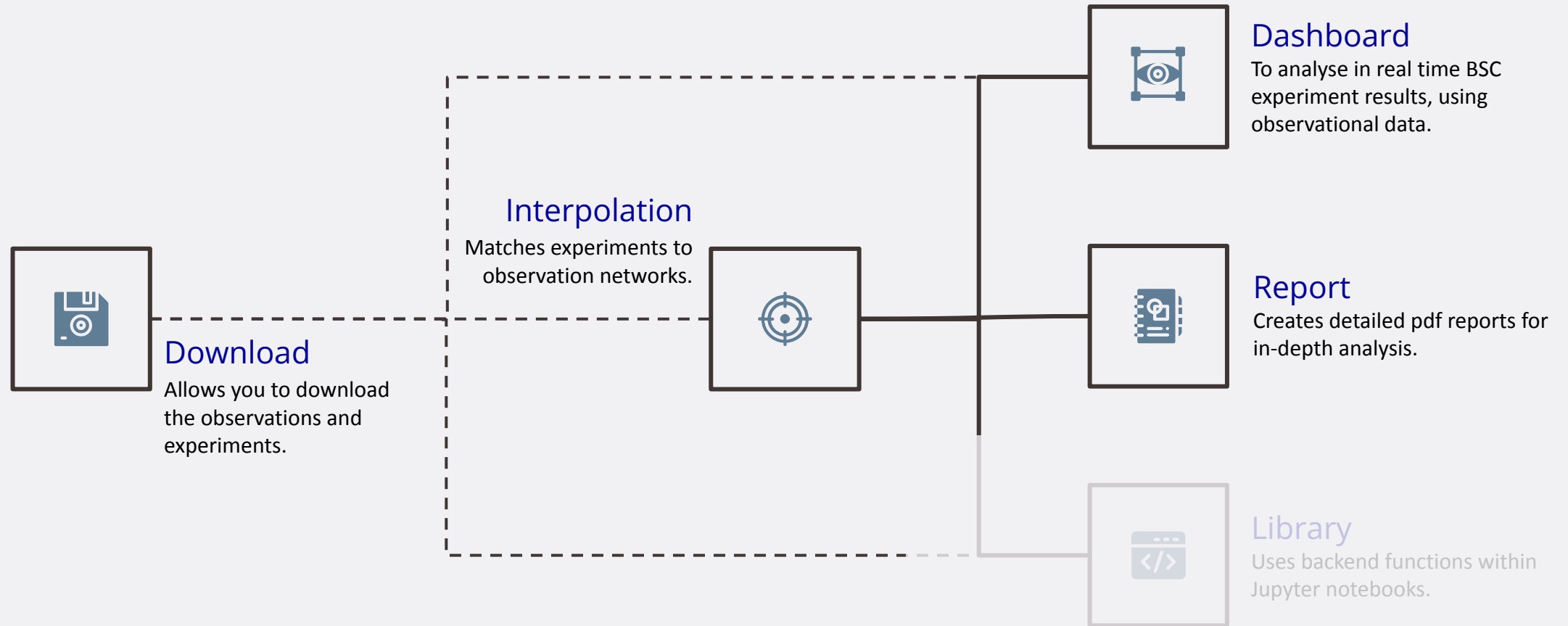
Structure



Dashboard



Structure



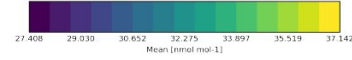
Report



Providentia R

Network : ['EBAS|sconco3']
 Species : ['sconco3']
 Temporal Resolution :
 Date Range : 20180101
 Experiments : ['EMEP', 'MONARCH', 'SILAM']
 Temporal Colocation :
 Spatial Colocation :
 Subsections : ['DEBUG-Europe']

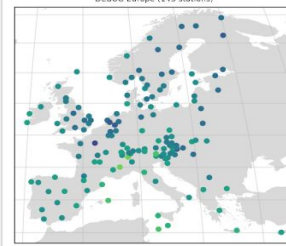
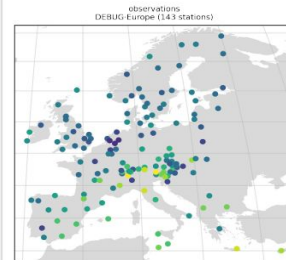
Heatmap Mean (Summary) multispecies



Map Median (Summary) EBAS|sconco3

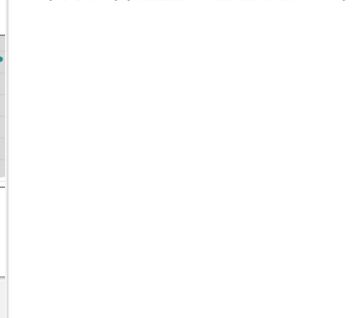
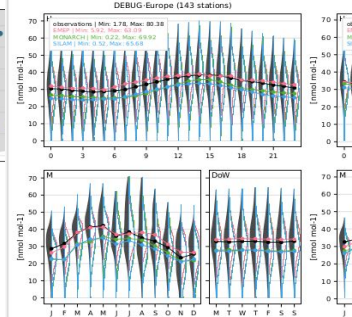


observations EMEP



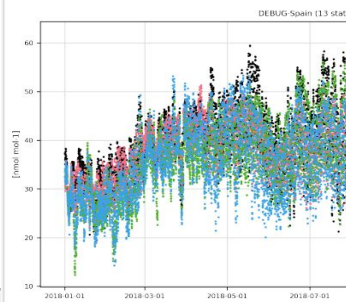
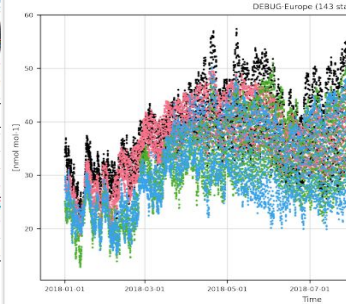
Violin (Summary) EBAS|sconco3

● observations ● EMEP ● MONARCH ● SILAM



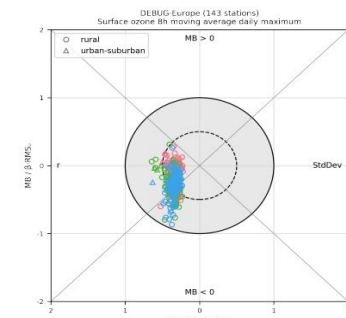
Timeseries (Summary) EBAS|sconco3

● observations ● EMEP ● MONARCH ● SILAM

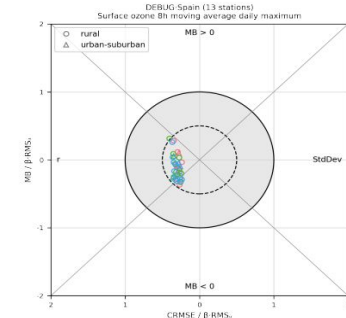


Fairmode target (Summary) EBAS|sconco3

● EMEP ● MONARCH ● SILAM

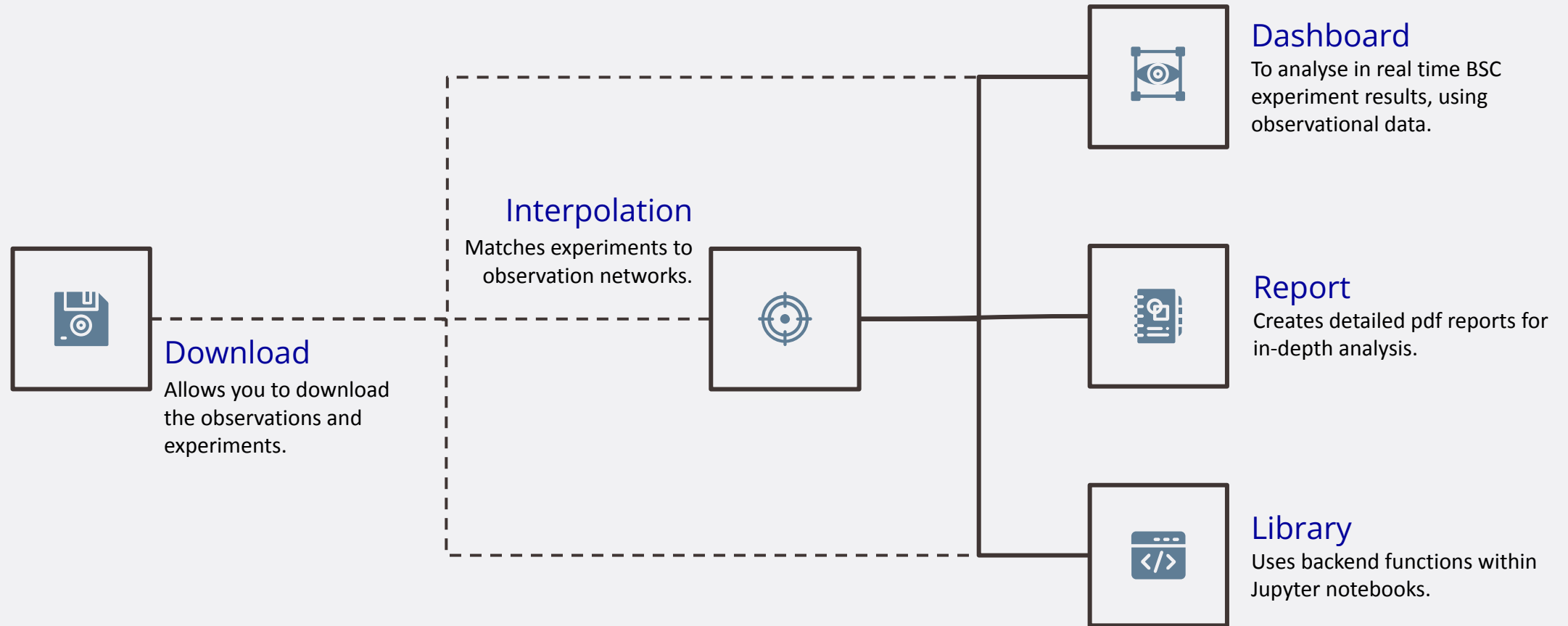


$\alpha=0.79$
 $\beta=2$
 $RV=60.12$
 $U_{95\%}=0.18$
 134 stations with coverage above 75%
 EMEP
 $MQ_{obs} = 0.50$
 0 stations with $MQ_i > 1$
 MONARCH
 $MQ_{obs} = 0.60$
 0 stations with $MQ_i > 1$
 SILAM
 $MQ_{obs} = 0.66$
 0 stations with $MQ_i > 1$



$\alpha=0.79$
 $\beta=2$
 $RV=60.12$
 $U_{95\%}=0.18$
 13 stations with coverage above 75%
 EMEP
 $MQ_{obs} = 0.41$
 0 stations with $MQ_i > 1$
 MONARCH
 $MQ_{obs} = 0.43$
 0 stations with $MQ_i > 1$
 SILAM
 $MQ_{obs} = 0.44$
 0 stations with $MQ_i > 1$

Structure

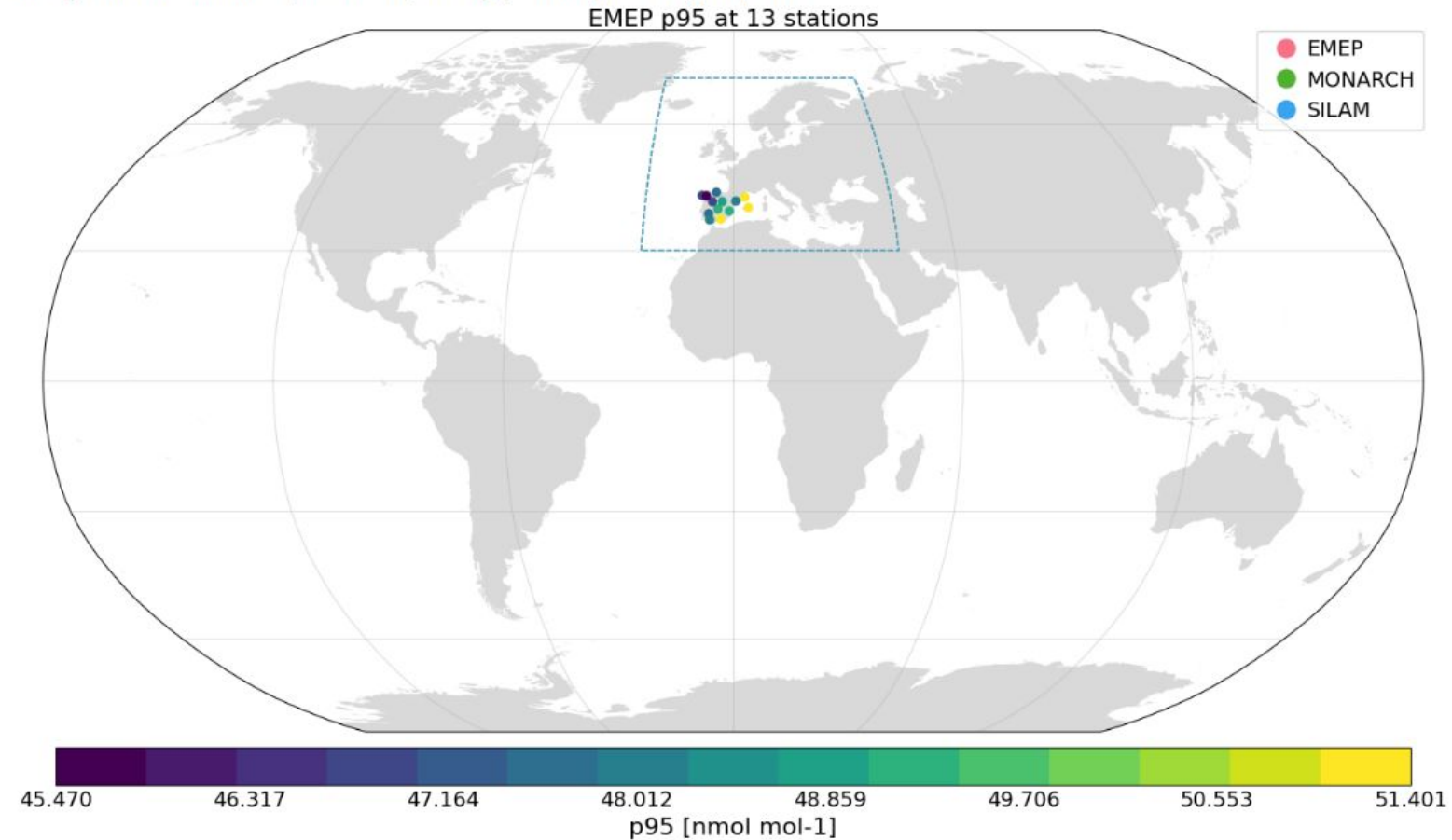


Library

```
import providentia as prv  provi = prv.load("interactive_template.conf")
```

```
# make a map plot (p95 statistic)  
provi.make_plot('map-p95', labela='EMEP', map_extent=[-180, 180, -90, 90], plot_options=['domain'])
```

Warning: Width and/or height have not been passed. The default values will be set.
Warning: More than 1 network or species defined, can only plot for 1 pair. Taking EBAS|sconco3.



ACTRIS

ACTRIS Configuration Setup

To use ACTRIS data in Providentia, you just need to set up a configuration file.

configurations/actris.conf

Section ([name])

```
[ACTRIS]
framework = actris
species = sconco3
resolution = hourly
start_date = 20180101
end_date = 20190101
experiments = cams61_chimere_ph2-eu-000 (CHIMERE)
temporal_colocation = True
spatial_colocation = True
report_type = standard
report_summary = True
report_stations = False
report_filename = PROVIDENTIA_Report
report_title = Report
```

Subsections ([[name]])

```
[[Barcelona]]
latitude = 39.8, 41.8
longitude = 1.5, 2.5
```

```
[[Spain]]
longitude = -9.39, 3.04
latitude = 35.945, 43.75
```

Download ACTRIS Data

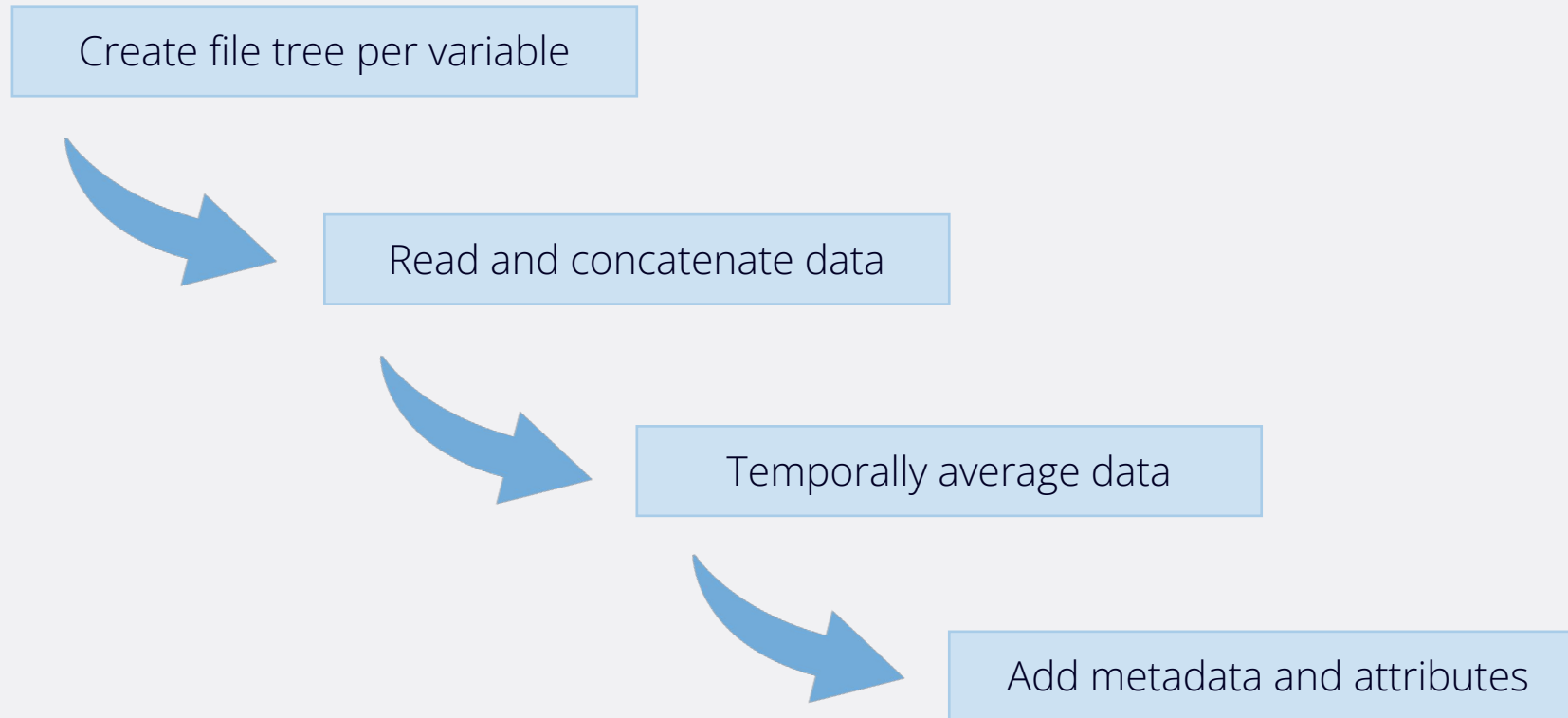
Using the download mode from Providentia, we can access Thredds (EBAS DOI in prod-actris-md.nilu.no) and retrieve the files to our machine in BSC format for the selected variable, resolution and period.

Once the config file is ready, downloading is simple. You run a single command:

```
./bin/providentia --config=your_configuration_name.conf --download
```

Providentia will automatically fetch the relevant ACTRIS data and begin formatting it.

Data Formatting



Demo

Launch the Tool

- Opening the dashboard:

```
./bin/providentia --config=your_configuration_name.conf
```

- Opening the jupyter notebook:

```
./bin/providentia --config=your_configuration_name.conf --notebook
```

- Creating reports:

```
./bin/providentia --config=your_configuration_name.conf --offline
```

- Interpolate the experiment to the observation:

```
./bin/providentia --config=your_configuration_name.conf --interpolation
```



Earth Sciences
Department



**Barcelona
Supercomputing
Center**
Centro Nacional de Supercomputación



Thank you for your attention!

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